Venkata Lakshmi Parimala Pasupuleti

AI Platform Engineer · TypeScript · LangGraph · Anthropic Claude · MCP · Tool Calling · HITL · LLM Evaluation · Agentic AI · OpenTelemetry · FastAPI · AWS

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PROFESSIONAL SUMMARY

AI Platform Engineer with 8+ years production SWE experience and 2 years hands-on agentic AI development. Built LeaseI: 8-agent LangGraph system with Anthropic Claude, MCP tool integration, durable state management, conditional routing, and HITL escalation — the same agent infrastructure Stord is building. Built AgentProbe: open-source AI evaluation infrastructure with trace-level observability, 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate), and CI-integrated regression harness. TypeScript + Python full-stack; strong on distributed systems, fault-tolerant event-driven architecture, OpenAPI design, and multi-tenant SaaS at scale. M.S. Data Analytics Engineering, George Mason University (GPA 3.87). — AgentProbe: github.com/venkatalakshmiparimala/AgentProbe

PROJECTS

Leasel: 8-Agent Agentic AI Platform with MCP + Tool Integration LANGGRAPH MCP ANTHROPIC CLAUDE

Python · TypeScript · LangGraph · MCP · Anthropic Claude · FastAPI · SQLite · OCR | github.com/venkatalakshmiparimala/Lease_complaince_rag

- **Agent infrastructure matching Stord's design exactly** — 8-agent LangGraph StateGraph (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation): each agent has scoped tool calls, durable memory via graph state, and conditional routing determining which agents to invoke per request type. MCP wires tool interfaces cleanly across agents. Confidence-based routing triggers HITL escalation when below threshold.
- Scope-aware tool systems exposing platform capabilities (document knowledge base, settings, historical incidents, regulatory corpus) to AI agents — same architecture as exposing orders/inventory to Stord agents. Fault-tolerant design: every agent path has defined fallback behavior.
- Eval: 93.6% precision, 87.9% F1 across 10 compliance rules on 1,240 documents; pass@3 88.3% via multi-trial sampling harness.

AgentProbe: Agentic AI Evaluation + Observability Infrastructure EVAL FRAMEWORK CI-INTEGRATED

Python · TypeScript · FastAPI · LangGraph · ChromaDB · GitHub Actions | github.com/venkatalakshmiparimala/AgentProbe

- Trace-level observability + 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate), LLM-as-judge scoring, CI-integrated regression harness with golden datasets, model routing across OpenAI/Anthropic/Groq/Ollama, offline + online evaluation loops. "Built the harness that tells you when your agents are working and when they're not." Used to evaluate LeaseI across 1,240+ compliance test cases.

WORK EXPERIENCE

Blue Horse Digital Sep 2026 – Present

AI/ML Engineer (Volunteer) · Remote

- Building ML models and data pipelines; surfacing insights from raw business data into application-facing APIs and reporting layers.

Magizhs Tech Solution May 2025 – Aug 2025

AI/ML Engineer Intern

- Built stateful agentic system (LangGraph + FastAPI): parse → plan → retrieve → validate → conditional routing. RAG over regulatory corpus; validation guardrails enforced in code before output. Shipped to real client; iterated on feedback weekly.

George Mason University Aug 2024 – Aug 2025

Research Associate

- Built ML pipelines (Python, R, TypeScript) across 190+ countries (R^2=0.977); fine-tuned HuggingFace classification model with PyTorch; deployed inference API; Isolation Forest anomaly detection across 3 research pipelines.

Entrata (PropTech / Real Estate SaaS) Sep 2018 – Aug 2024

Senior Software Engineer · Remote

- **8+ years production SWE** (Python, TypeScript, JavaScript, SQL) — owned compliance data pipelines, multi-tenant PostgreSQL schemas, and fault-tolerant async processing systems for 100K+ housing units. Production mindset: every release included validation, reconciliation, and rollback preparation.
- **Built fault-tolerant RabbitMQ event-driven system** — retry logic, idempotency guarantees, owned through production incidents; same stateful, high-availability service pattern as Durable Objects for agent persistence.
- **Designed OpenAPI-compliant REST services and integration contracts** — 20+ full-stack modules consumed by downstream services; multi-tenant data isolation, scope-based authorization across 100K+ units. Instrumented with OpenTelemetry; 200+ SQL queries optimized (30% gain).
- Shipped 20+ full-stack production modules across 120+ Agile sprints; microservices redesign (Docker) — 50% scalability improvement, 40% deployment time reduction.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur 2017

SKILLS

Agentic AI: LangChain, LangGraph, MCP (Model Context Protocol), Multi-Agent Orchestration, Tool Calling, Scope-Aware Tool Systems, Context Management, Durable State Management, Conditional Routing, HITL, Confidence-Based Routing, Agent Reasoning, Agent Memory, Prompt Engineering, LLM-as-Judge, Hallucination Detection, LLM Evaluation, Model Routing, LLMOps, Trace-Level Observability, Agent Tracing

Languages & Frameworks: TypeScript, Python (8+ yrs), JavaScript, Java, SQL, R, Bash — FastAPI, React.js, Django, REST APIs, OpenAPI

AI Models & SDKs: Anthropic Claude (AI SDK), OpenAI API, Groq, Ollama, HuggingFace, PyTorch, ChromaDB, Qdrant, Embeddings, RAG Pipelines, Hybrid Search, Reranking

Cloud & Infra: AWS (S3, EC2, RDS, Lambda, Step Functions, Bedrock), GCP, Docker, Kubernetes, Terraform, CI/CD (GitHub Actions), OpenTelemetry, Prometheus, Grafana, Serverless Architecture, Distributed Systems, Fault-Tolerant Systems, Multi-Tenant Architecture, Idempotency, PostgreSQL, SQLite, Redis, RabbitMQ, Apache Kafka